

- [Home](#)
- [What Is APS](#)
- [Articles](#)
- [Research](#)
- [Glossary](#)
- Domains ▾
 - [Conceptual Foundations](#)
 - [Evolutionary Dynamics](#)
 - [Developmental Organisation](#)
 - [Ecological Organisation](#)
 - [Cognition and Mind](#)
 - [Social Organisation](#)
 - [Diagnosis and Evidence](#)
 - [Philosophy of Biology](#)
 - [Methodology and Explanation](#)
 - [Artificial Systems and Boundaries](#)
-  [Search](#)

1. [Home](#)
2. / [Glossary](#)
3. / Ecosystem

Ecosystem

Canonical Glossary | [Ecology](#) | Created: 18 May 2026 | Revised: 2 October 2026

 **Canonically locked** since 18 May 2026 · Revised 2 October 2026.

Definition

An ecosystem is a distributed continuity organisation composed of interacting organisms, environmental processes, energetic flows, and ecological constraints through which viability-oriented persistence is sustained across scale and time.

In Brief

In APS, ecosystems are not merely collections of organisms within an environment, but distributed ecological continuity structures through which persistence, adaptation, perturbation, resilience, and environmental coupling become organised across interacting scales.

[Ecosystem](#)

[Definition](#)

In APS, an **ecosystem** is a distributed continuity organisation composed of interacting organisms, environmental processes, energetic flows, and ecological constraints through which viability-oriented persistence is sustained across scale and time.

Ecosystems are not merely collections of organisms occupying a shared environment.

They are recursively organised ecological continuity structures linking:

- persistence;
- perturbation;
- adaptation;
- resilience;
- environmental coupling;
- and multiscale biological organisation.

APS therefore approaches ecosystems as distributed forms of organised persistence.

APS Box — What Is Explanatory Grammar?

Explanatory grammar refers to the conceptual relations through which an explanatory approach identifies its questions, distinguishes relevant phenomena, and formulates candidate explanations.

In APS, this grammar begins from living organisation rather than from a privileged list of entities or causal levels. Agency, Process, and Scale provide complementary analytic projections of one viability-oriented, constraint-closed organisation: Agency concerns what living systems do, Process concerns how organisational continuity is maintained through change, and Scale concerns where persistence is organised across spatial and temporal extents.

Explanatory grammar can therefore shape what an inquiry makes salient without determining in advance what counts as an adequate explanation. APS may direct attention toward organisational dependencies, viability, continuity, constraints, and relations across spatial and temporal extents, but identifying those features does not by itself establish explanatory gain.

Whether an APS account contributes additional explanation must instead be assessed relative to a specified explanandum and the relevant established comparator. A mechanistic, evolutionary, developmental, ecological, or other explanation is not incomplete merely because it does not employ APS terminology or all three analytic projections. Where APS identifies a relevant dependency not captured by the comparator, it may add explanatory value; where it merely redescribes what is already explained, it does not.

In brief: explanatory grammar helps organise what an explanatory approach asks and makes salient; comparative assessment determines what the resulting explanation actually contributes.

Ecosystems and Organised Persistence

Living systems do not persist independently of ecological organisation.

Organisms depend continuously upon:

- energetic exchange;
- nutrient cycling;
- ecological interaction;
- microbial organisation;
- climatic stability;
- behavioural coordination;
- and environmental continuity.

Ecosystems therefore participate directly in the persistence conditions of living systems.

APS consequently treats ecosystems not as external environmental containers, but as continuity organisations within which viability-oriented persistence becomes distributed across ecological relations.

Ecosystems Are Not Static Containers

APS rejects the idea that ecosystems are static regions containing organisms.

Ecosystems are dynamic continuity structures continuously reorganised through:

- organism–environment coupling;
- energetic flow;
- perturbation;
- adaptation;
- ecological interaction;
- and evolutionary transformation.

Organisms simultaneously:

- depend upon ecosystems;
- reorganise ecosystems;
- and contribute to ecological continuity itself.

Ecosystems therefore emerge through recursively organised ecological activity rather than through fixed environmental boundaries alone.

Ecosystems and Environmental Coupling

Ecosystems depend upon environmental coupling.

Living systems remain continuously coupled to:

- atmospheric conditions;
- nutrient systems;
- microbial communities;
- trophic relations;
- climatic variation;
- and ecological feedback structures.

These couplings are reciprocal.

Environmental conditions constrain persistence possibilities, while living systems simultaneously modify ecological organisation through ongoing activity.

APS consequently approaches ecosystems as:

distributed coupling structures through which ecological continuity is sustained.

Ecosystems and Scale

Ecosystems operate across interacting scales.

Ecological organisation extends across:

- molecular processes;
- microbial communities;
- organisms;
- populations;
- landscapes;
- climatic systems;

- and evolutionary timescales.

No isolated scale fully explains ecosystem organisation.

For example:

- microbial activity shapes nutrient cycling;
- behavioural organisation alters population dynamics;
- climatic processes affect physiological viability;
- and evolutionary history constrains ecological resilience.

APS therefore approaches ecosystems through multiscale continuity analysis rather than through isolated local interaction alone.

Ecosystems and Perturbation

Ecosystems exist under continual perturbation.

Disturbance may involve:

- climatic fluctuation;
- habitat transformation;
- species migration;
- predation;
- disease;
- resource instability;
- or anthropogenic disruption.

Perturbation reveals ecosystem organisation.

Stable ecological conditions often conceal:

- dependency structures;
- resilience capacities;
- adaptive flexibility;
- and continuity vulnerabilities.

APS consequently treats ecosystems as dynamically reorganising continuity systems rather than static equilibria.

APS Box — Perturbation Reveals Organisation

In APS, perturbation can be diagnostically useful because disruption may make organisational dependencies visible that are difficult to distinguish under relatively stable conditions.

A system observed only during successful functioning may not reveal which relations contribute to maintaining that functioning. Perturbation can establish contrasts through which regulation, compensation, repair, reorganisation, dependence, and failure become empirically accessible.

Under disruption, a system may degrade, remain dependent upon external support, compensate partially, or reorganise its activity in ways that maintain or re-establish conditions required for continued functioning. Comparing these responses can reveal dependencies, regulatory capacities, limits of viability, and relations involved in maintaining living organisation.

Perturbation is therefore a useful investigative strategy rather than a universally privileged diagnostic method. Not every organisational relation requires perturbation to be identified, and disruption does not by itself establish

an APS explanation.

Where perturbation reveals that particular activities, constraints, or relations make a relevant difference to viability, those contrasts can provide evidence for an organisational explanation. Constraint closure, where relevant, concerns the mutual maintenance of constraints involved in sustaining living organisation rather than independence from environmental or material support.

Key Point: Perturbation can make organisational dependencies experimentally tractable by revealing how a living system responds when conditions affecting viability are altered. Whether those dependencies provide explanatory gain must still be established relative to the explanandum and relevant comparator.

Ecosystems and Resilience

Ecosystem persistence depends upon resilience.

Ecological resilience concerns the capacity of ecosystem organisation to:

- absorb disturbance;
- redistribute constraints;
- reorganise continuity;
- and preserve viability under changing conditions.

Resilience may involve:

- biodiversity;
- redundancy;
- adaptive flexibility;
- trophic reorganisation;
- developmental plasticity;
- or ecological succession.

APS therefore approaches ecosystem resilience as:

continuity-preserving ecological reorganisation across scale.

Ecosystems and Adaptation

Adaptation and ecosystems continuously co-organise one another.

Ecological organisation shapes:

- selection pressures;
- behavioural possibilities;
- developmental trajectories;
- and persistence strategies.

At the same time, organisms transform ecosystems through:

- niche construction;
- migration;
- cooperation;
- predation;
- ecosystem engineering;
- and environmental modification.

Ecosystems therefore evolve historically through ongoing reciprocal continuity transformation.

APS consequently rejects static conceptions of ecological organisation.

Ecosystems and Semiosis

Ecosystems are also semiotic organisations.

Living systems continuously respond to:

- ecological signals;
- environmental gradients;
- resource indicators;
- behavioural cues;
- and viability-relevant environmental differences.

Ecological continuity therefore depends partly upon meaningful environmental differentiation distributed across interacting systems.

For living organisms:

- food sources matter;
- predators matter;
- habitats matter;
- toxins matter;
- and social relations matter

because they affect persistence possibilities.

APS consequently approaches ecosystems not merely materially, but semiotically.

Ecosystems and Constraint Closure

Ecosystems extend continuity through distributed constraint relations.

Persistence depends upon recursively organised interactions distributed across:

- organisms;
- populations;
- energetic flows;
- environmental conditions;
- and ecological coupling structures.

Ecosystems therefore cannot be understood through isolated causal chains alone.

Ecological continuity emerges through distributed relations of reciprocal organisation across scale and time.

APS consequently approaches ecosystems as:

distributed persistence organisations structured through ecological continuity relations.

APS Reframing of Ecosystems

Many conventional approaches treat ecosystems primarily as:

- environmental regions;
- species assemblages;
- trophic networks;
- or resource systems.

APS reframes ecosystems organisationally.

Ecosystems concern:

- distributed persistence;
- ecological continuity;
- environmental coupling;
- perturbation;
- resilience;
- adaptation;
- semiosis;
- and multiscale ecological organisation.

This integrates ecosystems directly into the broader explanatory architecture of biology itself.

Summary

In APS, ecosystems are distributed continuity organisations linking organisms, environments, energetic flows, perturbations, and ecological constraints across scale and time.

Ecosystems:

- distribute persistence across ecological relations;
- reorganise continuity under perturbation;
- shape adaptation and resilience;
- and contribute directly to viability-oriented organisation.

APS consequently approaches ecosystems not as static environmental containers, but as:

distributed ecological continuity structures within organised persistence.

Referenced By

- [Resilience, Ecology, and Continuity](#) Article Canonical
- [The Ecological Organisation of Life](#) Article Canonical
- [Why Organisms Cannot Be Understood Apart from Their Environments](#) Article Canonical

Framework

- [What Is APS](#)
- [Understanding APS](#)
- [Core Structure](#)

Content

- [Articles](#)
- [Glossary](#)
- [Search](#)

Clusters

- [Conceptual Foundations](#)
- [Evolutionary Dynamics](#)
- [Developmental Organisation](#)
- [Ecological Organisation](#)
- [Cognition and Mind](#)
- [Social Organisation](#)
- [Diagnosis and Evidence](#)
- [Philosophy of Biology](#)
- [Methodology and Explanation](#)
- [Artificial Systems and Boundaries](#)

Research

- [APS Research](#)
- [Publications](#)
- [Citation Policy](#)
- [Versioning & Canonical Status](#)

Platform & Policies

- [About](#)
- [Purpose of This Platform](#)
- [Contact](#)
- [APS & AI](#)

APS Publishing Platform — A structured knowledge system for the APS Framework.